SearchChord: Prompt Preselection and Field Recomposition for One-Step Image Editing

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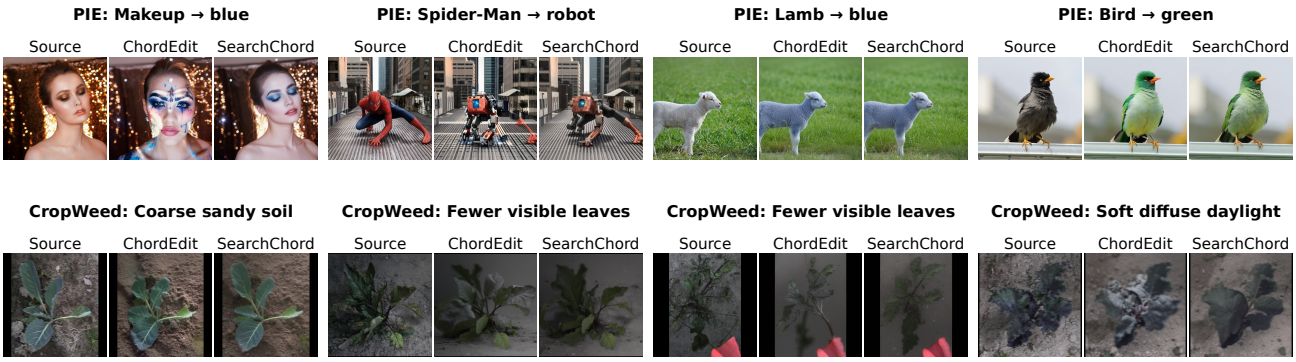


Figure 1. **Eight saved edits: four PIE-Bench and four CropWeed examples.** Sources are paired with ChordEdit (one field) and SearchChord (TextSim → EditReward; ten fields from 100 candidates), with common refinement.

Abstract

Effective prompt and seed search with one-step image editors requires balancing edit fidelity, source preservation, and computation. We present SearchChord, a training-free framework that combines text-based preselection, optional image verification, and latent-field recomposition. TextSim preselects ten of 100 candidates using text similarity, computes their control fields, and optionally reweights them using image-verifier scores. Experiments on PIE-Bench and CropWeed-EditBench reveal a consistent preservation–semantic trade-off: recomposition improves preservation of unchanged content, while candidate selection can achieve stronger target-text alignment. On PIE-Bench, diagnostic oracle recomposition reaches 24.73 dB PSNR, whereas oracle selection from 200 candidates reaches 25.46 CLIP-Edited. Practical verification partly recovers semantic alignment, but recomposition still loses edit fidelity, including on agricultural edits. These findings motivate text-first preselection and stronger verification for balancing preservation, edit fidelity, and cost. Project page.

1. Introduction

Text-guided image editing seeks to change requested attributes while preserving unrelated content. Recent one-step editors reduce the iterative cost of this task: SwiftEdit adapts one-step diffusion for fast editing [13], while ChordEdit uses training-free, inversion-free low-energy transport [10]. Editing-specific reward models, including EditReward and EditScore, complement these generators by evaluating generated edits [11, 21]. Together, these developments motivate studying how candidate generation and verification can balance preservation, edit fidelity, and computation.

Fast editors enable search over alternative prompts and random seeds, raising a central question: should we select one complete edit, combine candidate control fields, or preselect candidates before image generation? Stronger edits can disturb identity, geometry, and non-target content. Building on ChordEdit, SearchChord ranks prompts with TextSim and preselects candidates before computing fields, optionally verifies retained candidate images, and recomposes their fields before a common final generation step (Figure 2). Shared candidate pools and fixed refinement isolate recomposition from changes to the base editor. Figure 1 shows saved outputs.

Expanding the candidate pool offers more alternatives, but its benefit depends on candidate quality, selector relia-

bility, and evaluation cost [5, 15, 18]. We investigate three questions: whether text-based preselection identifies useful candidates before generation; how field recomposition compares with selecting one candidate from the same pool; and how these choices affect preservation and target-instruction alignment. Strong preservation may reflect a weaker edit, while stronger transformations can disturb non-target content. We therefore evaluate unchanged-region fidelity and target-text alignment separately on both benchmarks, using shared pools and fixed settings to assess trade-offs between quality and computation, and between preservation and semantic alignment.

Our main contribution is **SearchChord**, a training-free framework for candidate search and field recomposition in one-step image editing. TextSim preselects ten of 100 prompt/seed candidates before field computation; optional image verification then reweights the retained fields, which are combined before a common final generation step. The resulting pipeline integrates text-based candidate ranking, image-based assessment, and complete-field aggregation while retaining the base editor and refinement. With shared candidate pools, we compare single-candidate selection with equal- and score-weighted recomposition to isolate how these choices affect the final edit. Evaluations on PIE-Bench and CropWeed-EditBench, including separate weather, soil, and relative leaf-amount edits for 3,387 plant instances, characterize the resulting trade-offs between preservation, target-text alignment, and computation, including the semantic loss that practical image verification only partially recovers.

2. Related Work

Fast and training-free editing. Diffusion editing commonly uses inversion or iterative denoising to preserve source content [6, 12]. FlowEdit removes explicit inversion but integrates vector fields over multiple steps [8]. SwiftEdit and ChordEdit adapt editing to one-step generators [10, 13]. A subsequent ChordEdit analysis examines the roles of timestep selection, transport, and proximal refinement [9]. We retain the base estimator and refinement in the main search study; the matched-center controls isolate a separate estimator question.

Test-time scaling and verification. For LLMs, test-time scaling can improve outputs through additional inference compute: repeated sampling expands solution coverage, but imperfect verification can limit selection gains [2]; difficulty-aware search and revision improve budget use [18]; Archon optimizes combinations of inference techniques [17]. These findings motivate candidate diversity, reliable verification, and adaptive compute allocation as promising directions for image editing. Diffusion search [1, 24] similarly depends on reliable rewards [5, 23]. ELECT, ADE-CoT, and GridAR use early filtering, adaptive budgets, and pruning [7, 14, 15]. We combine fixed-budget text preselection, EditReward/EditScore verification [11, 21], and complete Chord-field recomposition.

Field composition. SplitFlow decomposes multi-attribute target prompts and combines sub-target flows through projection and aggregation [22]. We instead rank complete edit candidates for the same target and aggregate their Chord control fields before one final transport. This distinction concerns the candidate construction and composition unit, rather than the general idea of field aggregation.

Agricultural generative vision. Agricultural imagery varies with illumination, soil, growth stage, and season [19]. Generative augmentation can support recognition when outputs remain plausible and label-consistent [4, 20]. CropWeed-EditBench evaluates appearance and relative plant-structure edits in this setting. Its image-editing metrics may help assess the suitability of edited images for downstream classification or detection tasks, such as those created by inserting plant instances into a field background.

3. SearchChord

Overview and notation. A *candidate* pairs a target-prompt variant with a seed. Its *control field* is a Chord latent update; applying the update, refining, and decoding produces a *candidate image*. We consider N candidates and combine T fields into one final edit. Full-pool verification generates all N candidate images before retaining T; TextSim ranks all N texts and preselects T candidates before computing their fields.

3.1. Chord Candidate Fields

The source image is denoted by x, its description by c_s, and the target prompt by c_t. Candidate i uses prompt variant $\tilde{c}_{t,i}$ and seeded noise ϵ_i. The residual $\mathcal{R}_i(s)$ subtracts the source-conditioned clean-latent prediction from the target-conditioned prediction at the same noised source state and time s. It estimates the update associated with the requested change. Reusing candidate noise across times makes the temporal comparison use the same perturbation. The original Chord estimator combines two such residuals:

$$u_i = \frac{t\,\mathcal{R}_i(t-\delta) + \delta\,\mathcal{R}_i(t)}{t+\delta}, \tag{1}$$

with the published default $t = 0.90$, $\delta = 0.15$, and transport scale $\lambda = 1.0$ [10]. Unless otherwise stated, the main experiments also keep ChordEdit's proximal refinement fixed at $t_c = 0.30$ so every selection strategy receives the same final semantic refinement.

Exploring alternative field estimators. ChordEdit uses causal one-sided kernel smoothing [10]; we proposed Bi-Chord to explore symmetric smoothing to reduce potential

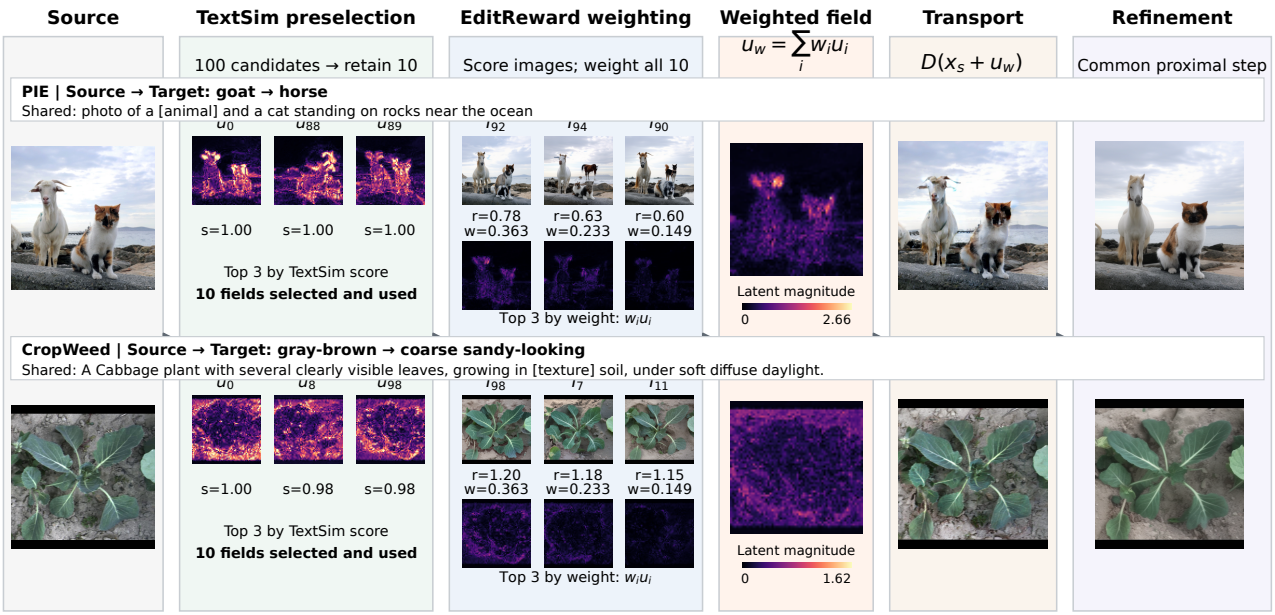


Figure 2. **SearchChord stage traces on PIE and CropWeed examples.** Prompt arrows show changed spans; bracketed slots retain shared wording. TextSim scores s_i preselect ten of 100 candidates; the top three fields are shown (ties by candidate index). Saved EditReward scores r_i weight all ten; the three largest weighted contributions are displayed. All ten fields enter the aggregate, without reselection. Heatmaps show field magnitudes and weighted contributions $w_i u_i$, with a common scale within each row. The combined field produces separate transport and proximal-refinement outputs. Intermediate images and fields are diagnostic CPU replays from saved prompts, original seeded CUDA noise, selections, and weights. Neither example appears in Figure 1; benchmark masks are used only for evaluation.

temporal bias. We compare the original asymmetric estimator with *BiChord*, which samples a symmetric temporal neighborhood,

$$u_i^{\mathrm{Bi}} = \alpha \mathcal{R}_i(t-\delta) + (1-2\alpha)\mathcal{R}_i(t) + \alpha \mathcal{R}_i(t+\delta), \quad (2)$$

with $\alpha = 0.25$. Its weights sum to one, $\alpha+(1-2\alpha)+\alpha = 1$, so no denominator is needed, whereas Chord normalizes by $t+\delta$. BiChord uses $t = 0.80$, $\delta = 0.15$, and timesteps $(0.65, 0.80, 0.95)$ within the diffusion interval. A Chord control at $t = 0.80$ isolates the estimator from the center-time change.

BiChord-RA makes the side weight risk-adaptive rather than fixed. Let σ_i^2 denote a local residual-variability estimate and C_i a local curvature estimate around t. We use

$$\alpha_i^{\mathrm{RA}} = \mathrm{clip}_{[0,\,1/2]}\left(\frac{2\sigma_i^2}{\delta^4 C_i^2 + 6\sigma_i^2}\right). \quad (3)$$

For each image, σ_i^2 averages the population residual variance across two noise draws and the three diffusion timesteps. The curvature term C_i^2 is the mean squared second difference of the noise-averaged residuals divided by δ^4, over latent channels and positions. Greater curvature reduces the side weight in Eq. 3, keeping the estimate closer to the center residual. Table 1 shares IDs, two noise samples, and metric denominators across all four estimators.

3.2. Candidate Construction

We explore candidate budgets from $N = 1$ to $N = 200$, using $N = 100$ as baseline experiments to balance diversity and computation. Each candidate pairs a prompt variant with a noise seed. FLAN-T5-base [3] generates distinct prompt variants intended to preserve the target meaning; we deduplicate each batch and fill shortfalls with the original target using fresh seeds. Larger pools retain earlier candidates for controlled comparisons, with their average composition shown in Figure 3(b). Because textually similar prompt variants may alter meaning, final refinement uses the original target prompt to reinforce the requested edit.

3.3. TextSim Preselection

TextSim assesses each prompt variant's alignment with the target before image generation. We average the generator text encoder's token features using the attention mask to exclude padding, and apply L2 normalization to the resulting representations. Let $e_s = f(c_s)$, $e_t = f(c_t)$, and $e_i = f(\tilde{c}_{t,i})$ be the resulting source, target, and candidate vectors.

The score combines two forms of alignment:

$$s_i^{\text{tar}} = \cos(e_i, e_t), \tag{4}$$

$$s_i^{\text{dir}} = \cos(e_i - e_s, e_t - e_s), \tag{5}$$

$$s_i^{\text{text}} = \alpha s_i^{\text{tar}} + (1-\alpha) s_i^{\text{dir}}, \qquad \alpha = 0.5. \tag{6}$$

Target alignment measures how closely a prompt variant matches the target description, while direction alignment measures whether the change from the source description aligns with the requested edit. Their equal-weight average accounts for both aspects. We retain the T highest-scoring candidates, compute their fields, and combine them using text-derived weights, as described below. For example, $(N, T) = (100, 10)$ ranks 100 candidates but computes only ten fields. Notably, TextSim operates solely on text: it cannot detect visual artifacts, distinguish different seeds paired with the same prompt, or guarantee that prompt variants preserve the intended meaning. Despite these limitations, we hypothesize that preselecting ten candidates from a pool of 100 and averaging their control fields can improve editing. We assess this potential through separate measures of source preservation and edit fidelity.

3.4. Coarse-to-Fine Verification Cascade

The cascade adds image evidence after text preselection. TextSim first retains the Top-T candidates from the N-candidate pool. We compute only those fields and generate their candidate images, then give the source–instruction–output triplets to EditReward or EditScore. The reported EditReward route is

$$\begin{aligned} 100 \text{ candidates} &\xrightarrow{\text{TextSim}} 10 \text{ fields}, \\ 10 \text{ fields} &\xrightarrow{\text{EditReward}} 10 \text{ weighted fields}. \end{aligned} \tag{7}$$

For weighted cascades, image scores replace the text-derived weights on the same ten fields. All ten contribute to the combined field; Best-of-10 instead returns one candidate. EditScore follows both procedures. This design tests whether image evidence improves the weighting of a fixed text-selected set. A verifier can observe artifacts that TextSim cannot, but it cannot recover a candidate already rejected by preselection. Figure 2 displays this sequence and its transport/refinement outputs.

3.5. Practical Verification and Diagnostic Oracle

Practical verifiers score source–instruction–output triplets without benchmark masks. Their scores estimate edit quality and instruction following. We use EditReward and EditScore; standalone CLIP image–text verification is not evaluated. CLIP-Whole/Edited below are evaluation metrics.

The oracle uses benchmark metrics to diagnose available candidate quality. Each metric ranks the same N candidates on $[0, 1]$: best is 1, worst is 0, and ties receive average ranks. PSNR and CLIP are maximized; MSE and LPIPS are minimized. Superscripts $-\text{MSE}$ and $-\text{LPIPS}$ denote these lower-is-better rankings, not negative error values.

We average the three preservation ranks and the two semantic ranks separately, then give these two averages equal weight. This avoids letting preservation dominate simply because it has three reported metrics:

$$\begin{aligned} S_i^{(5)} = \tfrac{1}{6} r_i^{\text{PSNR}} + \tfrac{1}{6} r_i^{-\text{MSE}} + \tfrac{1}{6} r_i^{-\text{LPIPS}} \\ + \tfrac{1}{4} r_i^{\text{CLIP-W}} + \tfrac{1}{4} r_i^{\text{CLIP-E}}. \end{aligned} \tag{8}$$

Preservation and semantics each receive 50% total weight. Rank normalization makes metric units comparable but describes relative standing within a pool. Undefined preservation terms are omitted and available weights renormalized. The oracle uses no target reference image; source-preservation and edit-mask information make it diagnostic only.

3.6. Top-T Field Recomposition

Given selector scores S_i, let $\mathcal{T}_{N,T}$ contain the T highest-scoring candidates among N. Best-of-N uses one field ($T = 1$). Recomposition instead combines the complete latent updates in that set:

$$u_{\text{mean}} = \frac{1}{T} \sum_{i \in \mathcal{T}_{N,T}} u_i, \tag{9}$$

$$u_{\text{w}} = \sum_{i \in \mathcal{T}_{N,T}} w_i u_i,$$

$$w_i = \frac{\exp(\tilde{S}_i/\tau)}{\sum_{j \in \mathcal{T}_{N,T}} \exp(\tilde{S}_j/\tau)}. \tag{10}$$

Mean aggregation assigns weight $1/T$; weighted aggregation assigns higher scores larger positive weights summing to one. Here $\tilde{S}_i$ is the oracle score in Eq. 8, or a normalized score rank for a practical selector. TextSim and full-pool verifiers rank within N; cascade verifiers rank only the T images they score. Temperature $\tau = 0.25$ controls concentration; smaller values favor the strongest candidates and larger values approach uniform weights.

We express the update relative to the original candidate field:

$$u_\beta = u_0 + \beta(u_{\text{agg}} - u_0). \tag{11}$$

Thus $\beta = 0$ uses the baseline field, $\beta = 1$ uses the aggregate, and intermediate values interpolate between them. The main comparison fixes $\beta = 1$; extrapolation with $\beta > 1$ is not established by these results. If z is the encoded source latent, transport gives $z' = z + \lambda u_\beta$. Common proximal refinement uses the original target prompt and base seed before decoding. Combining fields before image generation avoids pixel averaging and isolates transport changes.

3.7. Reading Quality–Compute Comparisons

Oracle Best-of-N describes available candidates under privileged scoring. Recomposition produces a new field, so we evaluate its generated output separately. A higher combined score need not improve every metric, and the oracle does not bound each ensemble-output metric.

Field count measures materialized Chord updates, excluding prompt generation, text encoding, verification, and image generation. Reducing 100 fields to ten lowers the candidate-field count by 90%; wall-clock savings also depend on these other stages. Table 3 reports their measured stage costs under explicit boundaries.

4. Experiments

4.1. Benchmarks and Evaluation Protocol

PIE-Bench. We use the 700-sample PIE-Bench benchmark for direct comparison with prior real-image editing work [6, 10]. Candidate 0 is the original target prompt with seed 42. The main comparison uses one baseline candidate or 100 search candidates and reports aggregation sizes $T = 1, 10$, and 100. TextSim emphasizes $(N, T) = (100, 10)$ versus full $(100, 100)$ field computation.

Metrics, masks, and denominators. We use the unchanged PIE evaluator with CLIP ViT-L/14 and LPIPS SqueezeNet. For PSNR, MSE, and LPIPS, the edited area is zeroed in both source and output before full-frame evaluation. CLIP-Whole uses the full output; CLIP-Edited zeros pixels outside the annotated edit region. The preservation metrics are undefined when the unedited fraction is below 0.001. They use the same 556 valid samples; both CLIP scores use all 700 common sample IDs. Missing preservation terms are omitted and remaining oracle weights renormalized. MSE and LPIPS are displayed multiplied by 10^3.

Masks are evaluation inputs, including to the diagnostic oracle. They do not constrain generation through masked blending or background freezing. Background masks define the evaluated edit region; they do not prevent plant changes.

CropWeed-EditBench. CropWeed-EditBench contains 3,387 deduplicated crop/weed bounding-box instances from top-down field images and bounding-box annotations collected by us. Each crop includes a 10% surrounding context margin and measures at least 300×300 pixels before resizing. Each image has three PIE-compatible source/target prompt pairs—weather, soil type, and relative leaf amount—sharing one source prompt. Local Qwen3-VL-2B-Instruct (4-bit) estimates source attributes; templates change one target attribute while retaining the others.

The three categories are evaluated separately across the 3,387 plant instances. Shared filtered coverage is 3,387 weather, 3,387 soil-type, and 3,386 relative leaf-amount specifications (10,160 in total). All methods use the same source images, prompts, and masks.

Table 1. Base-field estimator ablation on PIE-Bench. All four rows are rerun with two noise samples, seed 42, one edit step, $\delta = 0.15$, and proximal refinement at $t_c = 0.30$. Chord at $t = 0.80$ matches the center of BiChord and BiChord-RA. CLIP-Edited uses all 700 common samples; PSNR uses the same 556 samples with unedited fraction at least 0.001. These two-noise results are separate from the one-noise baseline in Table 3. Bold marks column-best metrics.

Field estimator	t	PSNR ↑	CLIP-Edited ↑
Chord (default)	.90	23.52	**21.85**
Chord (matched center)	.80	**24.37**	21.66
BiChord ($\alpha = .25$)	.80	23.71	21.67
BiChord-RA	.80	23.53	21.72

Automatic mask construction. We generate masks with SAM2.1 Hiera-base-plus [16] on an aspect-preserving working grid with longest side at most 768 pixels. RGB/HSV vegetation-color cues provide positive point prompts, while a central box spanning 8%–92% of each image dimension localizes the focal plant. Predicted mask quality, vegetation coverage, and color purity rank the box proposals. Additional point prompts cover vegetation outside the focal mask; accepted point masks, the focal mask, and vegetation-color seeds form a union that also protects neighboring plants during background edits.

Let P denote the focal-plant mask, V the vegetation union, and L the working image's shorter side. With D_r denoting dilation by an elliptical element of radius r (rounded to at least one pixel), the editable regions are

$$\begin{aligned} M_{\text{leaf}} &= D_{0.015L}(P), \\ M_{\text{weather}} = M_{\text{soil}} &= 1 - D_{0.006L}(V). \end{aligned} \tag{12}$$

Thus leaf edits include a small plant margin, whereas weather and soil share a background mask. Low-confidence or fallback cases are flagged for review; these automatic masks are not manually annotated segmentation ground truth.

Export alignment and interpretation. Masks return to native crop coordinates by nearest-neighbor resizing. Complete crops and masks are then jointly center-padded to squares and resized to 512×512, using Lanczos interpolation for RGB and nearest-neighbor for binary masks. This preserves aspect ratio without further cropping. Masks are encoded as zero-based, row-major PIE run-length pairs; decoded-mask round trips and image/mask alignment are checked for all exports, with representative overlays inspected. These checks establish geometric consistency, not pixel-level segmentation accuracy. Value 1 denotes editable pixels; black padding is 0 and contributes to preservation metrics without constraining generation. Leaf targets request more/fewer leaves; CLIP-Edited cannot establish exact count accuracy or biological

Table 2. Selector roles and information access. Relative costs are qualitative stage descriptions, not measured runtimes. TextSim is a pre-generation text-only preselector; EditReward and EditScore are practical image verifiers; the balanced Joint-5 oracle uses benchmark metrics and is diagnostic only. "Image?" indicates whether the edited candidate must already be generated.

Selector	Stage	Image?	Benchmark GT/mask?	Deployable?	Relative verifier cost	Main role
Joint-5 balanced	diagnostic	Yes	Yes	No	evaluation only	primary headroom estimate
EditReward	post-generation	Yes	No	Yes	High	edit quality + instruction following
EditScore	post-generation	Yes	No	Yes	High	visual quality + instruction following
TextSim	pre-generation	No	No	Yes	Very low	reduce field/image budget

Table 3. PIE-Bench quality–compute comparison with common proximal refinement and $\beta = 1$. Best-of-N delivers one scored candidate; Top-T recomposes fields from the same pools. TextSim Best-of-N computes one field; its ensembles and cascades compute ten. Cascade Best-of-10 selects one of those ten, while weighted cascades combine all ten. Stage times use a separate 20-image sequential benchmark defined below. Black bold marks overall column-best values; blue bold and teal bold mark the best within the full-verifier (EditReward/EditScore) and TextSim/cascade blocks, respectively, following the arrows and including displayed ties.

Method / selector	N	T	Fields	Unedited region			CLIP Semantics		Efficiency
				PSNR ↑	MSE$\times 10^3$ ↓	LPIPS$\times 10^3$ ↓	Whole ↑	Edited ↑	Mean stage (s) ↓
ChordEdit baseline	1	1	1	22.44	8.28	121.06	25.87	22.58	**0.49**
Joint-5 oracle: Best-of-N[†]	100	1	100	23.39	6.95	110.58	**27.93**	**25.22**	146.55
Joint-5 oracle: Mean Top-T[†]	100	10	100	24.72	5.08	91.02	24.52	21.13	146.98
Joint-5 oracle: Weighted Top-T[†]	100	10	100	**24.73**	**5.07**	**91.00**	24.52	21.17	146.74
Joint-5 oracle: Mean Top-T[†]	100	100	100	24.53	5.21	91.40	24.13	20.83	146.74
Joint-5 oracle: Weighted Top-T[†]	100	100	100	24.69	5.08	**91.00**	24.25	20.93	146.74
EditReward: Best-of-N	100	1	100	21.48	10.07	134.76	**26.32**	**23.08**	135.60
EditReward: Weighted Top-T	100	10	100	24.20	5.56	93.22	24.56	21.16	135.79
EditReward: Weighted Top-T	100	100	100	24.46	5.29	91.92	24.32	20.95	135.79
EditScore: Best-of-N	100	1	100	21.98	9.18	126.01	25.77	22.60	**109.46**
EditScore: Weighted Top-T	100	10	100	24.34	5.43	92.39	24.41	21.08	109.67
EditScore: Weighted Top-T	100	100	100	**24.49**	**5.25**	**91.65**	24.26	20.92	109.66
TextSim: Best-of-N	100	1	1	22.44	8.28	121.06	25.87	22.58	**0.55**
TextSim: Weighted Top-T	100	10	10	**24.39**	**5.41**	**92.67**	24.48	21.13	2.54
TextSim → EditReward: Best-of-10	100	1	10	22.13	9.00	126.92	**26.49**	**23.14**	13.80
TextSim → EditReward: Weighted	100	10	10	24.13	5.70	94.46	24.72	21.39	14.01
TextSim → EditScore: Best-of-10	100	1	10	22.30	8.63	124.37	25.98	22.63	11.02
TextSim → EditScore: Weighted	100	10	10	24.17	5.64	94.10	24.63	21.33	11.23

[†]Diagnostic oracle. Best-of-N retains its prompt/seed; recomposition uses the original target/base seed. Quality: 700 IDs, 556 valid preservation masks; EditScore duplicates checked against manifests/images. Sequential with unused scorers off and saved pools. Includes source encoding, field computation and image generation, selection, CPU oracle metrics, and verifier PNG/HTTP (EditReward batch 1; EditScore batch 4). Each variant pays the full shared pool/scoring cost. Verifier Best-of sums measured stages without recomposition; TextSim Best-of is newly timed. Excludes model loading, warm-up, prompt construction, input loading, and final evaluation/output writes; prepared-pool cost, not end-to-end latency. Separately, generating each 100-entry prompt/seed pool averages 29.09 s (FLAN-T5-base, FP32 CPU, eight threads; model loading and one warm-up pool excluded), including deduplication and target padding. This cost is needed once per $N = 100$ edit and excluded above; $N = 1$ uses the original target.

validity.

4.2. Main PIE-Bench Comparison

Table 3 compares single-candidate selection and recomposition for every selector. Best-of-N returns the highest-scoring candidate image with its own prompt and seed. Cascades select the best of the fixed TextSim top ten or combine all ten. Ties retain the earliest original candidate index.

Selection versus recomposition. Oracle Best-of-100 improves on candidate 0: PSNR rises from 22.44 to 23.39 dB and CLIP-Edited from 22.58 to 25.22. Weighted Top-10 further raises PSNR to 24.73 and lowers LPIPS to 91.00, but reduces CLIP-Edited to 21.17. Recomposition favors preservation rather than dominating selection. Mean and weighted Top-10 differ by only 0.01 dB; weighting alone does not explain the gain. Combining all 100 fields does not improve every metric.

Practical selection and recomposition. EditReward Best-of-100 reaches CLIP-Edited 23.08, above its weighted Top-10 (21.16), but PSNR falls from 24.20 to 21.48 dB. EditScore shows the same trade-off (22.60 versus 21.08 CLIP-Edited; 21.98 versus 24.34 dB). Both remain below the oracle's 25.22 semantic score. TextSim Best-of-100 reproduces candidate 0 on all 700 images: the unchanged target prompt attains maximal text alignment, so this selector cannot distinguish its seed alternatives. Its Top-10 recomposition instead

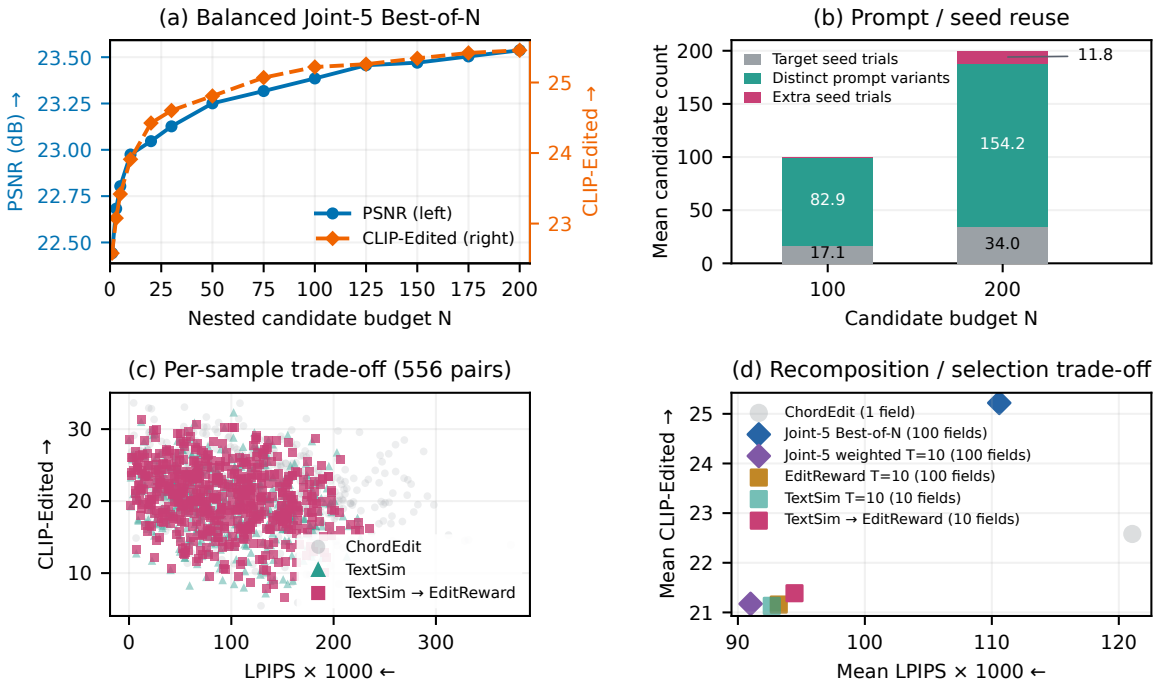


Figure 3. **Candidate budgets, prompt/seed reuse, and quality trade-offs.** (a) Measured Joint-5 budgets: 556 PSNR/700 CLIP-Edited IDs; separate axes. (b) Mean trials across 700 edits: target (gray), one per distinct non-original prompt variant (teal), and further trials with new seeds (rose). Reused variants count once in teal; their later trials enter rose. Bars count trials, not prompt sets. At $N = 100$, e.g., 83 variants plus 17 target trials give 100 candidates from 84 distinct prompts. (c) Identical-ID scatter. (d) $N = 100$ balanced Joint-5 comparisons. Prompt variants need not preserve meaning; Section 4.3 discusses seed-only results and attribution limits.

Table 4. Separate weather, soil, and relative leaf-amount evaluations for 3,387 CropWeed-EditBench plant instances. Shared coverage is 3,387 weather, 3,387 soil-type, and 3,386 relative leaf-amount specifications. ChordEdit uses $N = T = 1$; TextSim and its EditReward cascade use $N = 100$, $T = 10$. All methods share padded inputs, aligned masks, and fixed edit parameters. PSNR/MSE/LPIPS measure the unedited region; Whole/Edited are CLIP scores. Every shared mask is preservation-valid. Leaf edits are relative, without exact-count or downstream-recognition claims. Bold marks the best values within each family.

		Unedited region			CLIP Semantics	
Edit family	Method	PSNR ↑	MSE×10^3 ↓	LPIPS×10^3 ↓	Whole ↑	Edited ↑
Illumination/weather	ChordEdit baseline	25.87	3.07	95.87	**25.39**	**20.39**
	TextSim: Weighted Top-T	**26.53**	**2.65**	93.23	25.27	19.50
	TextSim → EditReward: Weighted	26.47	2.68	**93.07**	25.32	19.58
Soil appearance/type	ChordEdit baseline	25.24	3.62	103.71	**26.52**	**22.13**
	TextSim: Weighted Top-T	**25.85**	**3.16**	97.41	26.44	20.59
	TextSim → EditReward: Weighted	25.82	3.18	**97.31**	26.46	20.78
Relative leaf amount	ChordEdit baseline	25.39	3.56	119.52	25.32	**24.54**
	TextSim: Weighted Top-T	**25.50**	**3.52**	113.51	25.49	24.45
	TextSim → EditReward: Weighted	25.45	3.55	**112.44**	**25.53**	24.51

reaches 24.39 dB and CLIP-Edited 21.13.

TextSim → EditReward Best-of-10 reaches 23.14 CLIP-Edited and 22.13 dB, versus 21.39 and 24.13 for weighted recomposition. The EditScore cascade follows the same pattern (22.63/22.30 versus 21.33/24.17). Image verification improves selection over TextSim alone, while recomposition favors preservation. These fixed-pool means do not establish statistical significance; independent seed repetitions, confidence intervals, and human judgments remain absent.

Base-field controls. Table 1 uses two noise samples. Chord leads PSNR (24.37 at $t = 0.80$) and CLIP-Edited (21.85 at $t = 0.90$); neither BiChord variant surpasses these results, so SearchChord retains the published Chord default ($t = 0.90$).

4.3. Candidate Budgets and Quality Trade-offs

Figure 3(a) retains all $N \leq 100$ means and adds 70,000 saved candidates. Doubling N raises Joint-5 Best-of-N from 23.39 to 23.54 dB and 25.22 to 25.46 CLIP-Edited. Reranking only exact original-target seed trials gives 22.95/23.75 at $N = 100$ and 23.06/24.04 at $N = 200$, versus candidate 0's 22.44/22.58 (PSNR/CLIP-Edited). Seeds contribute to search gains; unequal subset sizes and unmatched seeds prevent attributing the full pool's remaining advantage to prompts alone.

Budget choice. Table 3 fixes $N = 100$ for controlled selector/recomposition comparisons and a pragmatic quality–cost trade-off. Larger pools are promising if added preparation, image generation, and scoring time is reduced and practical selectors realize the modest oracle headroom. $N = 200$ tests selection, not combining all 200 fields; optimality and saturation remain unestablished.

Per-sample and aggregate views. Panel (c) shows overlapping distributions on 556 identical IDs. Panel (d) retains $N = 100$ and the table's 556/700 denominators: oracle selection favors semantics, while recomposition favors preservation. Neither dominates every metric; prefix ranks can also yield non-monotonic metric trends.

Compute accounting. Table 3 reports 20-image mean stage times of 2.54 s for TextSim, 14.01 s for its EditReward cascade, and 11.23 s for its EditScore cascade. Runs are sequential on one RTX A6000, with only the required verifier loaded. Prepared-pool stages include source encoding and selection/scoring, but omit prompt construction and startup; they do not measure end-to-end speedups. A separate audit measured a peak of 5.77 GiB of generator-only PyTorch allocation on category-stratified PIE images (FP32, text batch 128), excluding external scorers, allocator reserve, and driver overhead. Fewer fields reduce computation; the resident generator and image generation determine peak allocation. Total deployed memory also depends on the verifier.

4.4. CropWeed-EditBench Evaluation

The transfer study compares three frozen methods: ChordEdit ($N = T = 1$), TextSim Weighted Top-10 ($N = 100$), and TextSim $\rightarrow$ EditReward Weighted Top-10 ($N = 100$). All use one noise sample, one edit step, seed 42, $t = 0.90$, $\delta = 0.15$, transport scale 1, proximal refinement at $t_c = 0.30$, and $\beta = 1$. The pipeline reads the same padded source images and filtered mapping for every method. The five metrics and mask conventions match the PIE evaluator; hyperparameters are not tuned by edit family.

Complete results. Table 4 uses independently audited shared IDs, metrics, outputs, and frozen protocol hashes. TextSim and its cascade share prompt/seed pools and selected indices. The final 2,285 cascade edits reuse prompt strings but recompute scores, fields, images, and metrics; only preparation cost changes.

Preservation and edit fidelity. TextSim improves mean PSNR over ChordEdit by 0.66 dB for weather and 0.62 dB for soil, while lowering CLIP-Edited by 0.90 and 1.54 points, respectively. For leaf amount, its PSNR gain is 0.11 dB and CLIP-Edited falls by 0.09. The EditReward cascade recovers 0.09, 0.19, and 0.06 CLIP-Edited points relative to TextSim, with PSNR lower by 0.06, 0.03, and 0.05 dB. It lowers LPIPS relative to TextSim in each family but remains below the baseline in CLIP-Edited. Thus image weighting shifts the operating point without resolving the semantic loss from recomposition. These descriptive means do not establish statistical significance or biological validity. Edits from the same plant are correlated, so uncertainty analysis should group them by source instance.

Figure 1 shows four visually inspected CropWeed examples from saved common outputs: soil changes, diffuse lighting, and relative foliage reductions. They are selected illustrations, without exact-count annotations or evidence of aggregate transfer gain.

Interpreting agricultural edits. Weather/soil transformations evaluate background edits and plant preservation; leaf transformations evaluate plant edits and preservation outside the plant. Cross-family scores therefore depend on mask area and task meaning. Black padding contributes to preservation scores: the shared protocol supports method comparisons, but scores are not equivalent to an unpadded or cropped benchmark.

An additional reward model can observe candidate images, but generic edit quality need not measure agricultural plausibility. Relative foliage changes may alter morphology or species cues even when CLIP alignment increases. Downstream recognition, exact counting, and label consistency need independent annotations and evaluation. We do not infer any of those outcomes from these five image-editing metrics.

5. Conclusion

SearchChord separates text preselection, optional image verification, and field recomposition. TextSim reduces a 100-candidate pool to ten materialized fields; image verifiers then select from or reweight the retained set. PIE-Bench oracle selection exposes semantic headroom, whereas combining Top-10 fields favors preservation. For weather, soil, and relative leaf-amount edits evaluated separately on 3,387 CropWeed plant instances, TextSim preserves unchanged content better but lowers CLIP-Edited; EditReward partially recovers alignment. These choices shift the preservation–semantic operating point. End-to-end efficiency and independently judged edit correctness remain to be evaluated.

References

[1] Gholamali Aminian, Idan Shenfeld, Amir Reza Asadi, Ahmad Beirami, and Youssef Mroueh. Best-of-n through the smoothing lens: Kl divergence and regret analysis. In *International Conference on Learning Representations*, 2026. 2

[2] Bradley Brown, Jordan Juravsky, Ryan Ehrlich, Ronald Clark, Quoc V. Le, Christopher Ré, and Azalia Mirhoseini. Large language monkeys: Scaling inference compute with repeated sampling. *arXiv preprint arXiv:2407.21787*, 2024. 2

[3] Hyung Won Chung, Le Hou, Shayne Longpre, Barret Zoph, Yi Tay, William Fedus, Yunxuan Li, Xuezhi Wang, Mostafa Dehghani, Siddhartha Brahma, Albert Webson, Shixiang Shane Gu, Zhuyun Dai, Mirac Suzgun, Xinyun Chen, Aakanksha Chowdhery, Alex Castro-Ros, Marie Pellat, Kevin Robinson, Dasha Valter, Sharan Narang, Gaurav Mishra, Adams Yu, Vincent Zhao, Yanping Huang, Andrew Dai, Hongkun Yu, Slav Petrov, Ed H. Chi, Jeff Dean, Jacob Devlin, Adam Roberts, Denny Zhou, Quoc V. Le, and Jason Wei. Scaling instruction-finetuned language models. *Journal of Machine Learning Research*, 25(70):1–53, 2024. 3

[4] Boyang Deng and Yuzhen Lu. Weed image augmentation by controlnet-added stable diffusion for multi-class weed detection. *Computers and Electronics in Agriculture*, 232: 110123, 2025. 2

[5] Leo Gao, John Schulman, and Jacob Hilton. Scaling laws for reward model overoptimization. In *International Conference on Machine Learning*, 2023. 2

[6] Xuan Ju, Ailing Zeng, Yuxuan Bian, Shaoteng Liu, and Qiang Xu. Pnp inversion: Boosting diffusion-based editing with 3 lines of code. In *International Conference on Learning Representations*, 2024. 2, 5

[7] Joowon Kim, Ziseok Lee, Donghyeon Cho, Sanghyun Jo, Yeonsung Jung, Kyungsu Kim, and Eunho Yang. Early timestep zero-shot candidate selection for instruction-guided image editing. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 18844–18854, 2025. 2

[8] Vladimir Kulikov, Matan Kleiner, Inbar Huberman-Spiegelglas, and Tomer Michaeli. Flowedit: Inversion-free text-based editing using pre-trained flow models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 19721–19730, 2025. 2

[9] Minghan Li, Jeremy Moebel, and Mengyu Wang. Rethinking one-step image editing through chordedit: Reproduction, simplification, and new insights. *arXiv preprint arXiv:2606.14042*, 2026. 2

[10] Liangsi Lu, Xuhang Chen, Minzhe Guo, Shichu Li, Jingchao Wang, and Yang Shi. Chordedit: One-step low-energy transport for image editing. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14398–14407, 2026. 1, 2, 5

[11] Xin Luo, Jiahao Wang, Chenyuan Wu, Shitao Xiao, Xiyan Jiang, Defu Lian, Jiajun Zhang, Dong Liu, and Zheng Liu. Editscore: Unlocking online rl for image editing via high-fidelity reward modeling. In *International Conference on Learning Representations*, 2026. 1, 2

[12] Ron Mokady, Amir Hertz, Kfir Aberman, Yael Pritch, and Daniel Cohen-Or. Null-text inversion for editing real images using guided diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6038–6047, 2023. 2

[13] Trong-Tung Nguyen, Quang Nguyen, Khoi Nguyen, Anh Tran, and Cuong Pham. Swiftedit: Lightning fast text-guided image editing via one-step diffusion. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 21492–21501, 2025. 1, 2

[14] Joonhyung Park, Hyeongwon Jang, Joowon Kim, and Eunho Yang. Progress by pieces: Test-time scaling for autoregressive image generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026. 2

[15] Xiangyan Qu, Zhenlong Yuan, Jing Tang, Rui Chen, Datao Tang, Meng Yu, Lei Sun, Yancheng Bai, Xiangxiang Chu, Gaopeng Gou, Gang Xiong, and Yujun Cai. From scale to speed: Adaptive test-time scaling for image editing. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 23272–23282, 2026. 2

[16] Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, et al. SAM 2: Segment anything in images and videos. *arXiv preprint arXiv:2408.00714*, 2024. 5

[17] Jon Saad-Falcon, Adrian Gamarra Lafuente, Shlok Natarajan, Nahum Maru, Hristo Todorov, Etash Kumar Guha, E. Kelly Buchanan, Mayee F Chen, Neel Guha, Christopher Re, and Azalia Mirhoseini. An architecture search framework for inference-time techniques. In *Proceedings of the 42nd International Conference on Machine Learning*, pages 52475–52507, 2025. 2

[18] Charlie Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar. Scaling llm test-time compute optimally can be more effective than scaling model parameters. In *International Conference on Learning Representations*, 2025. 2

[19] Daniel Steininger, Andreas Trondl, Gerardus Croonen, Julia Simon, and Verena Widhalm. The cropandweed dataset: A multi-modal learning approach for efficient crop and weed manipulation. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 3729–3738, 2023. 2

[20] Da Tan, Michael Beck, Christopher P. Bidinosti, Robert H. Gulden, and Christopher J. Henry. Generative diffusion models for agricultural ai: Plant image generation, indoor-to-outdoor translation, and expert preference alignment. *Computers and Electronics in Agriculture*, 249:111862, 2026. 2

[21] Keming Wu, Sicong Jiang, Max Ku, Ping Nie, Minghao Liu, and Wenhu Chen. Editreward: A human-aligned reward model for instruction-guided image editing. In *International Conference on Learning Representations*, 2026. 1, 2

[22] Sung-Hoon Yoon, Minghan Li, Gaspard Beaudouin, Congcong Wen, Muhammad Rafay Azhar, and Mengyu Wang. Splitflow: Flow decomposition for inversion-free text-to-image editing. In *Advances in Neural Information Processing Systems*, 2025. 2

[23] Zhuohao Yu, Steven Wu, and Adam Block. From curiosity to caution: Mitigating reward hacking for best-of-n with

pessimism. In *International Conference on Learning Representations*, 2026. 2

[24] XiangCheng Zhang, Haowei Lin, Haotian Ye, James Y. Zou, Jianzhu Ma, Yitao Liang, and Yilun Du. Inference-time scaling of diffusion models through classical search. In *International Conference on Learning Representations*, 2026. 2